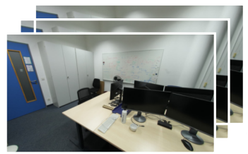
GeometryGS

SfM-anchored spatial Gaussian generation

TokenGS-scale backbone · 4,096 scene latents · 4,096 GS tokens · 1,024-d features

Pretrained · New · SfM · Optional

Input views



RGB + known cameras $\{I_v, P_v\}$

Same input views

PRETRAINED IMAGE–SCENE PATH

Multi-view image encoder

RGB + ray embeddings

12 layers · multi-scale

Scene latent aggregator

12 layers · 4,096 latents

Track-sampled image features

Known-pose SfM

Match · triangulate

Filter + FPS

SfM memory

$S = \{\mathrm{PE}(a_j), f_j, q_j\}$

3D · features · quality

SfM scene fusion

Cross-attn + FFN

2 residual blocks

Identity initialization

Scene memory $\widetilde{Z}$

GS token initialization

Query $\widetilde{Z}$; predict free centers

K **fixed** · $4096 - K$ **free**

$T_i = (h_i, c_i, r_i, t_i)$ · $K \le 1024$

Spatial GS states

Spatial GS decoder

4 layers · 1024-d

Cross-attention to $\widetilde{Z}$

Self-attn + relative 3D bias

FFN + free-center update

3D PE · radius · type

Local Gaussian head

Child MLP · bounded scales

$\mu_{ik} = c_i + r_i\,\mathrm{Ball}(\delta_{ik})$

64 Gaussians / token

$\mathcal{G}$

Gaussian rasterizer

RGB $\hat{I}$ + depth $\hat{D}$

TRAINING ONLY

RGB + laser depth / normals

Privileged supervision

Rendering losses

RGB + SSIM + depth + visibility

Optional soft chain · off by default

Same-surface links: $\rho\left(n_{ij}^{\top}(p_i - p_j)\right)$ · preserve tangential spacing

Solid arrows: inference data flow Dashed arrows: training supervision Gaussian glyphs: schematic